WMDR2-core

Package in package 'Model'

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WMDR2-core diagram

Class diagram in package 'WMDR2-core'

WMDR2core
Notes
1. temporalExtent shall be realized as time:interval
2. temporal<Name> elements other than temporalExtent shall be realized as an array
the values of <Name>, e.g., temporalTerritory = [[2000-01-01, 2024-07-13],
["https://codes.wmo.int/wmdr/territory/<ISO1>", "https://codes.wmo.int/wmdr/terri

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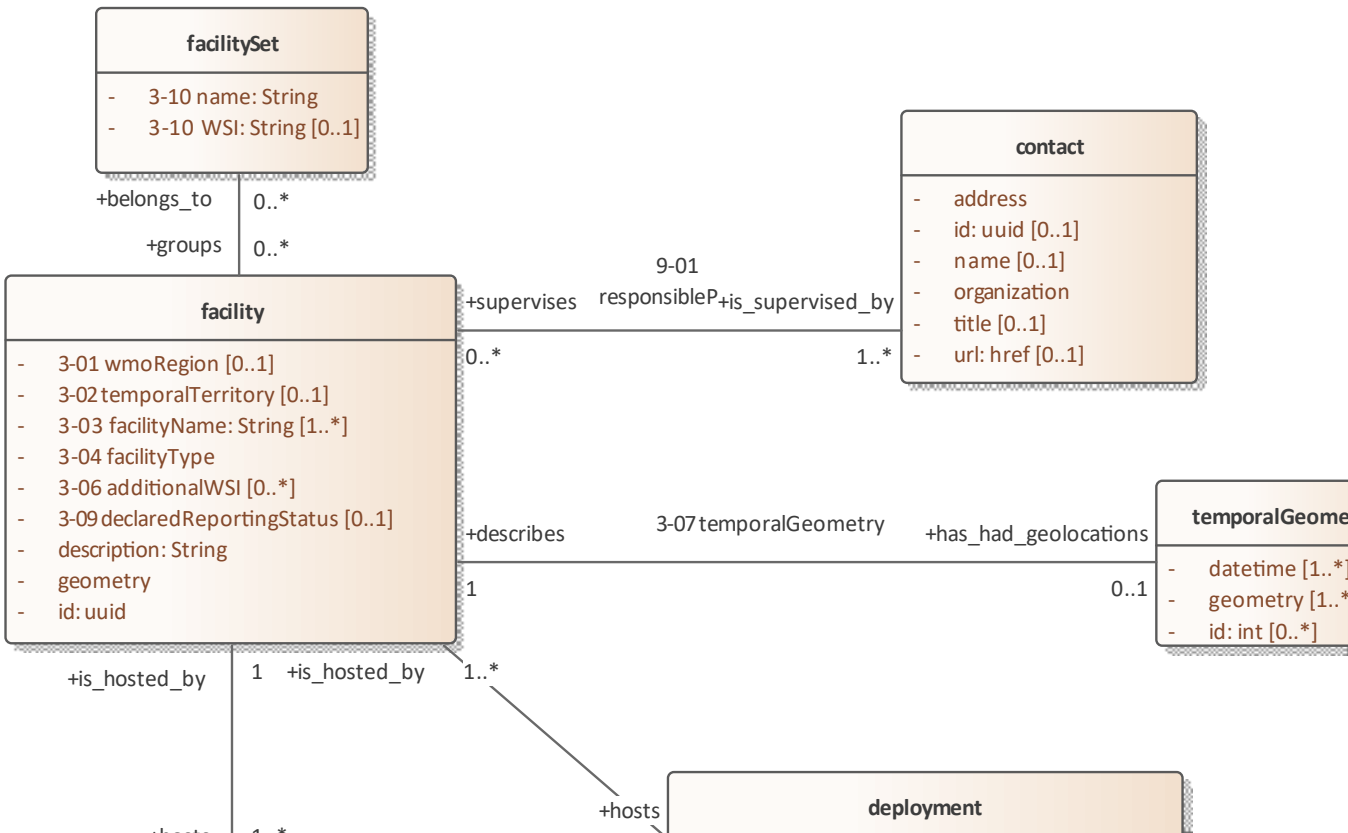

Class in package 'WMDR2-core'

impkement according to VCard 4.0 specification [0].

[0] https://datatracker.ietf.org/doc/html/rfc6350#page-23

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ATTRIBUTES

address : Private

ADR: The structured type value consists of a sequence of
address components. The component values MUST be specified in
their corresponding position. The structured type value
corresponds, in sequence, to
the post office box;
the extended address (e.g., apartment or suite number);
the street address;
the locality (e.g., city);
the region (e.g., state or province);
the postal code;
the country name (full name in the language specified in
Section 5.1).

Example:
ADR;TYPE=work:;;Federal Department of Home Affairs FDHA Federal Office of M
eteorology and Climatology MeteoSwiss Measurements and Data\nOperation Cen
ter 1 (OPC 1)\nP.O. Box 257;Zürich-Flughafen;;CH-8058;Switzerland

[Is static

id : uuid Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static

name : Private

ATTRIBUTES

url : href Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static

ASSOCIATIONS

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public
Cardinality

deployment

Class in package 'WMDR2-core'

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ATTRIBUTES

1-03 temporalExtent : Private
Alias: records:interval
[Is static

1-04 temporalSpatialExtent : Private
Alias: records:geometry

Typical spatial geo-referenced volume covered by the observations.

EXAMPLES:
(a) Air temperature at a surface observing site: Sydney Airport, New South Wales, Austra
alt. 6.0 m above mean sea level (amsl);
(b) The projected area or volume of the cone around a particular weather radar with a ma

ATTRIBUTES

5-02 observingMethod : Private

[Is static

5-05 heightAboveLocalReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static

5-05 local ReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface

[Is static

5-09 instrumentManufacturer : Private

[Is static

5-09 instrumentModel : Private

[Is static

5-09 temporalInstrumentSerialNumber : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static

5-12 temporalGeometry : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

- Geospatial location of instrument/sensor

[Is static

6-07 temporalDiurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES

temporalOfficialStatus : Private

[Is static

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public
Cardinality

ObservingProc

Association (direction: Unspecified) 7-03 temporalReportingSchedule

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public
temporalRepo
Cardinality

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public
Cardinality

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public
Cardinality

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1..*]

Target: Public
Cardinality

ATTRIBUTES

3-02 temporalTerritory : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities

[Is static

3-03 facilityName : String Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

[Is static

3-04 facilityType : Private

[Is static

3-06 additionalWSI : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

An array of well-formed WSIs

[Is static

3-09 declaredReportingStatus : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Condition: must be provided if no deployment is specified; otherwise: Optional, inferred
temporalReportingStatus + temporalProgramAffiliation.
Exception: "closed" over-rules deployment, in which case all deployments should be term

[Is static

description : String Private

[Is static

geometry : Private
Alias: 3-07 current geolocation

[Is static

ASSOCIATIONS

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) facility
Cardinality: [0..*]

Target: Public
Cardinality

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public
temporalGeom

Association (direction: Unspecified)

Source: Public groups (Class) facility
Cardinality: [0..*]

Target: Public
Cardinality

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1..*]

Target: Public
Cardinality

facilitySet

Class in package 'WMDR2-core'

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ATTRIBUTES

3-10 name : String Private

[Is static

Class in package 'WMDR2-core'

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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static

manufacturer : Private

This should come from a registry/catalogue, but should accept any organization.
[Is static

model : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

This should come from a registry/catalogue, but can accept "custom-built".
[Is static

ASSOCIATIONS

Association (direction: Unspecified) 5-09 instrument

Source: Public describes (Class) deployment
Cardinality: [0..*]

Target: Public
Cardinality

observation

Class in package 'WMDR2-core'

ATTRIBUTES

- observableProperty: a state variable (e.g., temperature), or another property (e.g., conce
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (ra
humidity (range), etc., as relevant for the result of the observation. Constraints shall be do
where the names shall come from a code list, and the values shall come from a code list d
text.

[Is static

1-01 observedGeometryType : Private

[Is static

1-01 observedVariableConstraints : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

observedVariable should be modeled using the I-ADOPT ontology approach. Combinatio
- observableProperty: a state variable (e.g., temperature), or another property (e.g., conce
- objectOfInterest: e.g., dry air (of the atmosphere)
- observedVariableConstraint(s): e.g., a size fraction, a specific wavelength, a pressure (ra
humidity (range), etc., as relevant for the result of the observation. Constraints shall be do
where the names shall come from a code list, and the values shall come from a code list d
text.

[Is static

1-03 temporalExtent : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

aggregate derived from deployments if deployments are provided; otherwise, specify beg

[Is static

10-01 contact : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ogc:contact includes "role"

[Is static

2-02 temporalProgramAffiliation : Private

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) observation
Cardinality: [1..*]

Target: Public
Cardinality

Association (direction: Unspecified)

Source: Public is_hosted_by (Class) facility
Cardinality: [1]

Target: Public
Cardinality

procedure

Class in package 'WMDR2-core'

Extension point to document
- ObservingProcedure
- (Raw)DataProcessing
- ...

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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) deployment
Cardinality: [1..*]

Target: Public
Cardinality

ObservingProc

ATTRIBUTES

id : int Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)
[Is static

ASSOCIATIONS

Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public describes (Class) facility
Cardinality: [1]

Target: Public
temporalGeom
Cardinality

temporalReportingSchedule

Class in package 'WMDR2-core'

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ATTRIBUTES

id : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
[Is static

interval : Private

An array of arrays of begin(inclusive) and end(exclusive) datetime stamps. If the end is n
open-ended. Intervals can overlap to indicate multiple schedules operating in parallel.
[Is static

schedule : Private

temporalGeometry

InformationItem in package 'WMDR2-core'

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ASSOCIATIONS

Association (direction: Unspecified)

Source: Public (Class) facility

Target: Public
temporalGeom